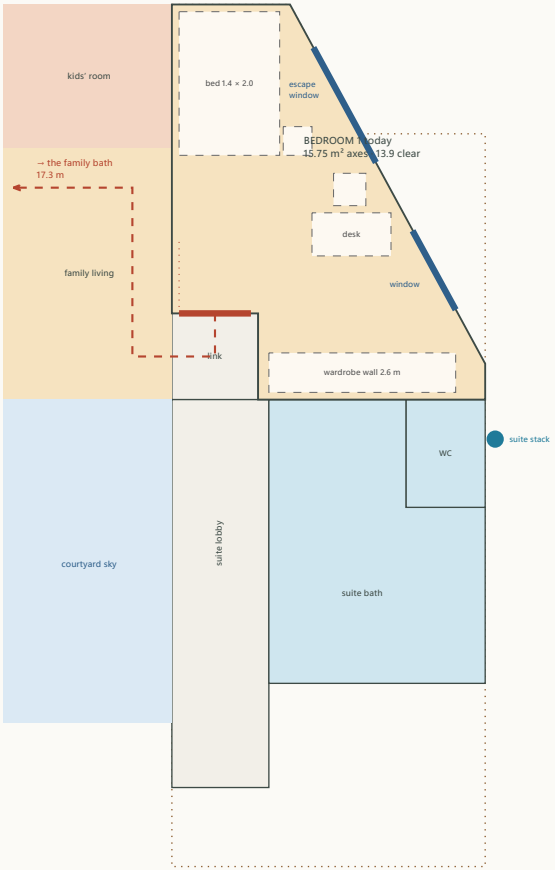
Bedroom 1's bathroom · design review F-05 · accept the route (A) or a private bathroom (B)? · Taif / Palm House · A ADOPTED · B NOT ADOPTED: young children's room (owner, V9.0)

OPTION A ADOPTED by the owner (2 Nov 2026): no private bathroom; the wet areas stay grouped in the two clusters; night lights along the route. 2 Nov 2026, after the stair flip (option B, F-b) · bedroom 1 is the north-east room over the maid wing, 15.75 m² on axes, 13.9 m² clear, one room without a bath · nothing in the design files changes until the owner decides · all sizes assumed

bedroom 1 · A · 1 m = 0.95 cm · north up · brown dotted = the ground floor below



bedroom 1 · B2 · 1 m = 0.95 cm · north up · brown dotted = the ground floor below



Option A · accept the route

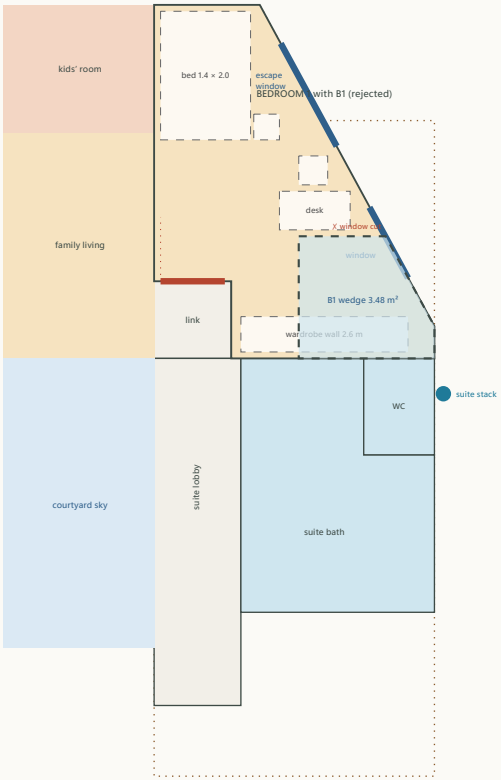
- **The walk, measured after Part 1** (the stair flipped, the family bath enlarged northwards; its door stays at x 4.25, y 9.3–10.3): bedroom 1's door → the family link → the family living, behind the sofa → the gallery's north part, past the stair arrival → the gallery → the bath's door: **17.3 m** (the same as before the flip: the bath grew, its door did not move). Five spaces, two of them lived in (the living, the gallery).
- **Who shares the family bath:** bedroom 1 (17.3 m), bedroom 2 (2.7 m), the kids' room and the family living (a bedroom later); bedroom 3 and the suite have their own. Since F-b it has **two compartments** (a WC by the shaft and the shower + double basin), so two children at once.
- **Comfort:** a child or a guest in bedroom 1 crosses the **family living** at night and in the morning, in sight of whoever is there; 17 m is the longest bathroom walk in the house (the suite's is 0, bedroom 3's 2.7, bedroom 2's 2.7).
- **Cost:** nothing; the plumbing stays as it is.

Option B · a private bathroom: the positions tried

- **B2 (the best, drawn):** a bath along bedroom 1's **south wall**, x 10.6–13.311, y 5.0–6.5, its east end following the cut wall: **3.79 m² on axes, 2.49 × 1.3 clear less the slant ≈ 3.0 m²**: a walk-in shower 0.9 × 1.2 at the east end (its corner 0.02 m west of the cut wall), the WC in the middle facing north, the basin on the west wall, a **sliding 1.0 m door** from the room (no swing), a **small window on the cut wall** (y 5.35–5.85, sill +1.60, obscured). It is **next to the master suite's wet stack**: the pipes run 2.5 m east in the bath's own floor zone to the suite's stack outside the east wall (y 7.05), **over the GF maid room only** (the kitchen and its hood zone start at y 7.0; nothing over the master bedroom).
- **B1 (the south-east wedge, dashed on page 2):** x 11.2–13.311 on the cut wall, 3.48 m²: a wedge, its north wall cuts the second window (y 4.15–5.25) and leaves 0.8 m of the wardrobe wall: worse than B2 for the same plumbing. **B3 (the north-east corner):** the north-east corner (x 10.6–12.0, y 1.0–2.9) holds the room's **escape window** and the bed's head wall: a bath there moves the escape window and the bed; rejected without drawing.
- **Bedroom 1 with B2: 11.96 m² on axes, ≈ 10.3 m² clear** (was 15.75 / 13.9): the bed 1.4 × 2.0 stays on the west wall with its bedside table, the desk and chair under the escape window, the wardrobe wall (2.6 m) becomes **two wardrobes, 1.3 m on the bath's wall and 0.9 m on the west wall** (2.2 m in all); the door's swing stays clear; the **escape window stays the room's** (the fire study unchanged); the second window (y 4.15–5.25) lights the desk. Still a full bedroom (a 1.4 bed, a desk, 2.2 m of wardrobe), no longer a two-bed room.

Plumbing, structure and windows for B2 · B1 for the record · the comparison · recommendation · the decision needed

bedroom 1 · B1 · 1 m = 0.95 cm · north up · brown dotted = the ground floor below



B2 · plumbing, structure, windows

- **Drain:** the shower, the WC and the basin to one branch running east along the bath's south side to the **suite's stack** outside the east wall (y 7.05), 2.5 m; the branch crosses the room's south wall line at x 13.311 and turns south outside the wall for 0.6 m (or a second stack beside the suite's; the MEP engineer). **Over the GF maid room only**: not over the kitchen (y ≥ 7.0), its hood zone (the east wall y 10.2–11.2) or the master bedroom (x 8.95–13.311, y 11.9–19.25).
- **Structure (`studies/structure.typ`):** the maid wing's first-floor bay is a **solid 0.20 slab** spanning east–west 4.36 m to the cut wall; a 0.15 depressed zone cannot be cut from 0.20. Two ways, the engineer's choice: **(i)** cast the bay 0.35 thick under the bath with its top depressed 0.15 (as the suite bath, X10), or **(ii)** keep the slab and run the pipes **below** it in the maid room's ceiling void (the maid room gets a gypsum ceiling anyway; 3.1 m clear left) with a wall-hung WC and a shower tray raised ≈ 0.10. The bath's walls are light partitions on the slab (no beam; B6 at y 6.5 is the room's south wall beam, untouched).
- **Windows:** the bath's small window on the cut wall faces the lane side (obscured glass, a vent); the room keeps both its windows. **Ventilation:** the window plus an extract fan to the roof riser is not available here: an extract through the cut wall (a small louvre) or a duct in the ceiling void to the east wall.
- **Neighbours:** the maid room below hears the bath (pipes in a ceiling void: wrap them); the suite's WC closet shares the stack; the family link and the suite lobby are untouched; the family bath keeps serving bedroom 2, the kids' room and the living.

Comparison

Criterion	A · accept the route	B2 · a private bath along the south wall
Comfort at night, privacy	17.3 m through the family living and the gallery, in sight of the living	0 m; a guest never crosses the family's rooms
Room size left	15.75 m² axes · 13.9 clear; 2.6 m of wardrobe; a two-bed room possible	11.96 m² axes · ≈ 10.3 clear; 2.2 m of wardrobe (two pieces); one bed + desk
Plumbing	none	2.5 m to the suite's stack, over the maid room only; a depressed or thickened bay, or pipes in the ceiling void
Structure	none	the maid wing's 0.20 solid bay: thicken to 0.35 (depressed top) or keep and raise the tray; light partitions
Windows	unchanged	a small obscured window on the cut wall; the room keeps both windows and its escape casement
Cost	nothing	one more small bathroom (fittings, tiling, waterproofing, the slab detail, the stack connection, a vent)
Effect on the neighbours	the family bath carries four users (two compartments since F-b)	the maid room's ceiling carries the pipes (noise: wrap); the family bath carries three users
Who it suits	bedroom 1 as a child's room in a family that shares baths	bedroom 1 as a guest room or an older child's room; resale value

Recommendation: B2, if bedroom 1 is to be a guest room or an older child's room; otherwise A

- The route is the longest in the house and crosses the living: that is the real cost of A, and the only one. B2 answers it for ≈ 3 m² of the room and one more wet room; it is the only tidy position (B1 is a wedge that cuts a window, B3 takes the escape window). **Conditions for B2:** the engineer chooses the slab detail (i) or (ii); the MEP engineer the stack connection and the vent; the wardrobe split accepted.
- **If A:** keep the family bath as F-b (two compartments) and note in the house rules that bedroom 1's user has the family bath; nothing else to do.

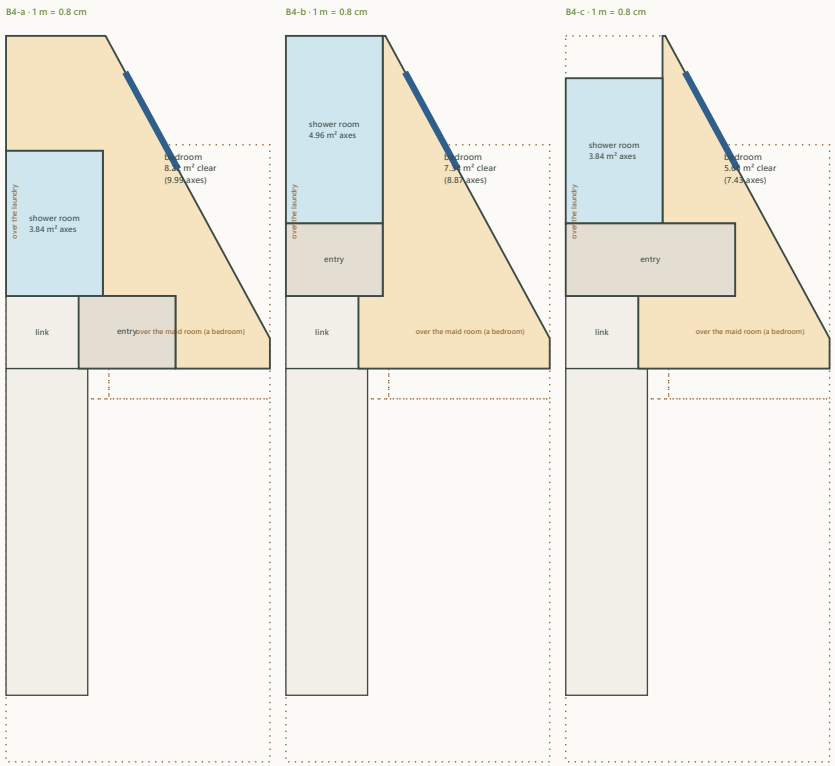
Decision needed from the owner

1. **A** (accept the 17.3 m route, no bath) or **B2** (the private bath along the south wall, ≈ 3.0 m² clear, the room 10.3 m² clear)? If B2: the slab detail is the engineer's; the wardrobe split (1.3 + 0.9 m) accepted?

Study checks passed (8)

Checks: B2 lies inside bedroom 1 (its vertices on or inside the room's polygon) · B2 leaves bedroom 1 ≥ 10 m² clear (11.96 m² on axes) · B2's fixtures do not overlap and stay inside its clear box · B2's shower corner stays west of the cut wall (x 12.8 ≤ 12.82 at y 5.1) · bedroom 1's furniture after B2: inside the room, not overlapping, the door's swing square clear · B2's drain runs over the GF maid room only (never the kitchen's hood zone or the master bedroom) to the suite's stack 2.5 m away · the escape window stays bedroom 1's in B2 (it is north of the bath) · B1's north wall (y 4.6) cuts the second window (y 4.15–5.25): a conflict to report. Nothing here changes the room CSVs, the design files or the structure; every size is assumed.

B4 (3 Nov 2026) · an en-suite shower room in a “hotel layout” for bedroom 1 · studied, NOT applied: the acceptance criteria 2 and 3 fail in every variant · NOT ADOPTED: young children’s room (V9.0)



Acceptance criterion	B4-a	B4-b	B4-c
(1) over wet / service rooms only (the laundry, the maid bath), short runs in the east cluster	OK	OK	OK
(2) the bedroom a usable rectangle ≥ 10 m² clear with its furniture	NO	NO	NO
(2’) … a clean rectangle	NO	NO	NO
(3) the two doors side by side on one straight line; the bed not seen from the door	NO	NO	OK
(4) the escape window kept in the bedroom	OK	OK	OK
bedroom clear m² (axes)	8.22 (9.99)	7.34 (8.87)	5.66 (7.43)
shower room axes m² · entry axes m²	3.84 · 1.92	4.96 · 1.92	3.84 · 3.36
(5) structure, the suite bath, the link, the gallery	OK	OK	OK

Clear ≈ axes less 0.10 m along every edge. (1) with a short new vertical in the laundry’s corner (a service room of the east cluster) to the buried east branch: the suite bath’s stack is 3 m away across the maid room’s ceiling, so the brief’s “east stack or suite stack” is met only by that new drop (acceptable: no new cluster). The bed can be hidden from the door in B4-a / B4-b (the door looks at the entry zone).

Why it fails: the room is a wedge

- Bedroom 1 is 15.75 m² on axes, ≈ 13.98 m² clear, **wide in the south (x 8.95–13.3 at y 5.3–6.5) and narrow in the north** (the cut wall runs from (10.59, 1.0) to (13.31, 6.0)). The brief’s programme is a 3.84 m² shower room + a ≈ 1.9 m² entry + a ≥ 10 m² clean rectangle = ≥ 15.7 m² with their walls: **more than the room holds**. Whatever is cut off for the shower room and the entry, the bedroom left is the wedge: 5.7–8.2 m² clear and never a clean rectangle.
- **Criterion (3)** (the two doors on one continuous line) needs the shower room and the bedroom side by side north of the entry’s wall; the only wet rooms below are the laundry (x ≤ 10.65) and the maid bath, so the shower room must sit at x ≤ 10.65 and the bedroom door east of it on the same line: that is B4-c, whose entry (2.8 m long) cuts the room in two. B4-a and B4-b keep a usable bedroom shape but their two doors meet at a corner.

The best variant, if the owner still wants an en-suite here: B4-b

- **B4-b:** the existing door from the link (y 5.3) opens into a 1.6 × 1.2 **entry zone**; straight ahead the **shower room** 1.6 × 3.1 on axes (4.96 m²; a level shower 1.0 × 1.4, the wall-hung WC and basin on a service wall on its east side x 10.55, the channel along it; mechanically vented: no outer wall of its own) **over the maid bath and the laundry only**, its drains down a new short vertical in the laundry’s north-east corner (a service room) to the buried east branch; the **bedroom** through a door on the entry’s east wall: 7.34 m² clear, an L with the bed 1.4 × 2.0 along the south wall (x 11.5–12.9, its head on the wall y 6.5), the desk under the escape window on the cut wall, a 1.8 m wardrobe on the service wall’s bedroom side (x 10.55, y 1.3–3.1); the bed is out of sight from the link’s door (it looks at the entry and the shower room’s door).
- **What it costs:** criterion (2) (≈ 7.34 m² instead of 10, an L) and criterion (3) (a corner, not one line). Everything else passes. **Alternative:** keep option A as decided on 2 Nov (no bath; since 3 Nov 2026 the first-floor family bath is a WC only, so bedroom 1 would have **no shower on its floor**: it would use bedroom 2’s or 3’s en-suite, or the suite’s); or **B2 of 2 Nov** (3.0 m² along the south wall), which fails criterion (1) (over the maid room).
- **Decision needed:** B4-b with its two failed criteria accepted, or another answer for bedroom 1’s shower (e.g. a shower added to the first-floor family WC: the strip y 7.9–9.2 is 2.75 × 1.3, too small for a shower without taking bedroom 2’s bath back).

B4 checks passed (4)

B4 checks: the room’s clear area (13.98 m²) cannot hold a 3.84 m² shower room + a 1.9 m² vestibule + a 10 m² bedroom (17.34 m² with their walls): the sum exceeds the room · every B4 variant fails at least one criterion (none is applied) · B4-a and B4-b keep the shower room over the laundry / maid bath only; B4-c (the doors on one line) is also over wet rooms only but its bedroom is two pieces · the bedroom proper stays under 10 m² clear in every variant (8.22 / 7.34 / 5.66 m²). Nothing here changes the room CSVs or the design files (B4 not applied); every size assumed, the plumbing to be designed by the plumbing engineer.

- **Not squeezed:** no variant was made to pass by shrinking the shower room below 1.6 × 2.4, the wardrobe below 1.8 m or the door swings.